

Atlantic Sea Scallop Backup Approach

Patrick J. Sullivan, Cornell University

Jonathon Peros, NEFMC

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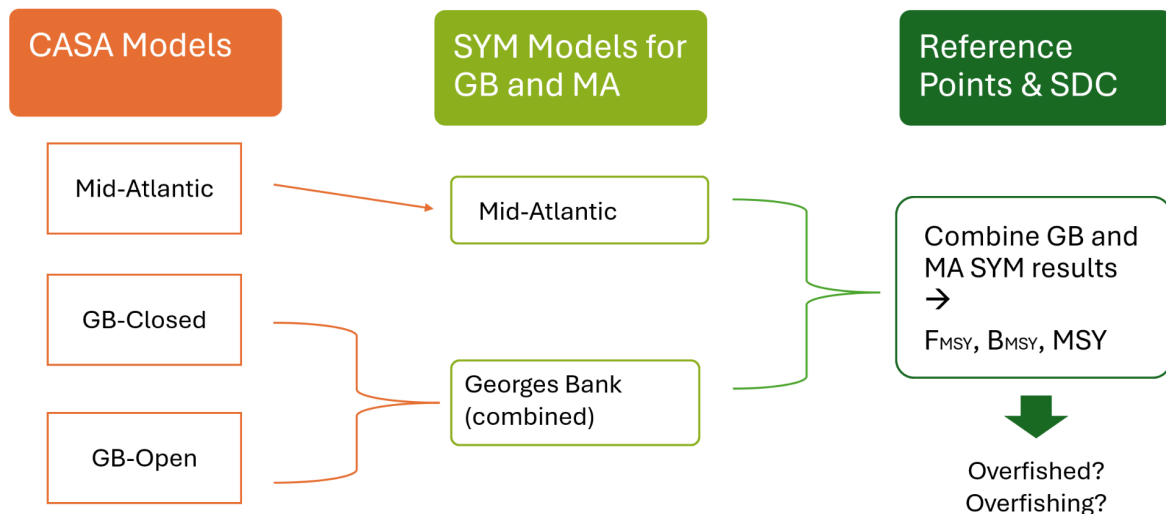
Introduction

This working paper addresses Term of Reference 8 for the Atlantic Sea Scallop Research Track Working Group (WG):

Develop a backup assessment approach to providing scientific advice to managers if the proposed assessment approach does not pass peer review or the approved approach is rejected in a future management track assessment.

The WG identified the parts of the process where a potential failure to achieve an acceptable result might occur (Figure 1).

Steps for Development of Reference Points and SDC



1. The Catch at Size Analysis model (CASA) fails to perform for any of the three management areas. The CASA model estimates the number of fully recruited scallops in the population for each management region (Mid-Atlantic, Georges Bank closed, and Georges Bank open).
2. The Stochastic Yield Model (SYM) fails to perform for any of the two areas (Mid-Atlantic or the combined Georges Bank area). The SYM combines the Mid-Atlantic

and Georges Bank results to derive a stockwide result. Using a Monte Carlo simulation, candidate reference points are provided, including recommendations for biological reference points of FMSY, BMSY, and MSY. SYM accounts for uncertainty in the CASA model estimates and addresses the additional issues of unaccounted mortality during the fishing process (e.g. bycatch mortality of discards and incidental mortality caused by gear on bottom). The Fmsy from the SYM model is used in status determination.

We note that a failure in any one of the three components of Part 1, or any of the two components of Part 2 will affect the calculation of biological reference points that are used in conjunction with the New England Fishery Management Council's harvest control rule to develop Over Fishing Limit (OFL) and Acceptable Biological Catch (ABC) estimates. Broadly speaking, any failure would require a backup plan, but the approach to remedy the situation would vary for each part of the process.

Methods

CASA Backup

The scallop dredge surveys are designed in such a way that the number of scallops observed by area of bottom covered by a tow and corrected for catchability can provide a measure of average scallop density by area. This density by area can be expanded by strata (e.g. SAMS area) to get the number of scallops by strata for each SAMS management area. These population estimates can be fed into the SYM to calculate reference points.

Similar calculations can be made using optical survey methods (See TOR 3 working paper).

SYM Backup

The SYM was outlined in an ICES peer reviewed journal article (Hart 2013) and has not, in substance, changed since its use in 2014 or during the 2018 Stock Assessment Workshop (65th SAW), when it was approved for use. The research track working group is recommending changes to M and recruitment in SYM in this assessment.

Reference points could be calculated deterministically using means or medians or trimmed means of the various CASA output parameters rather than through stochastic simulation. However, it is known that stochastic estimates of F_{MSY} are lower than deterministic estimates because only the stochastic reference points take into account the risk that F_{MSY} has been overestimated due to parameter misestimation. So conservative use of such reference points is recommended.

Model Free Backup

Should the failure of approval for any or all of the model steps in Parts 1 and 2 occur in such a way that status determination remains unknown there are other options that can be considered. For example, the scallop dredge surveys or the optical surveys could still be used to determine

standing stock biomass. In this scenario, a conservative determination of fishing mortality would be determined by using the realized catch over the last five years along with an effort determination from the LPUE to come up with an appropriate Acceptable Biological Catch (ABC) for the fishery. However, under such a scenario the productivity of the stock may remain unknown and consequently certain status determination metrics such as the Overfishing Limit (OFL) would remain unavailable.

Discussion

These backup plans are only for consideration when the main assessment approach, CASA, is unable to converge in any formulation, is not completed, or is rejected during peer review of the research track or management track process. The WG has provided these backup plans taking into consideration possible reasons why this might happen. There are also some reasons this backup approach might not work.

As with any other assessment, a more practical concern is that the approaches for all three assessment regions rely on the scallop dredge surveys and possibly the optical surveys. If these surveys are unable to be completed successfully in each region, the approaches could fail. Missing surveys are a different problem from zero captures during a survey. Obviously, as the amount of zero captures increases, so does the risk to the stock. Methods exist to fill in the gaps under certain sets of assumptions regarding total coverage by the existing survey methods.

Conclusions and Recommendation

The working group recommends that if a backup plan is needed, the adjusted area survey measures (e.g., dredge, drop camera, HABCAM) of fully recruited scallop density expanded by survey strata (e.g., SAMS areas) and then summing over strata can be used to arrive at an estimate of scallop abundance. This abundance estimate can be combined with similar estimates for the other management region or with the CASA estimates from each region to be processed by the SYM to arrive at the appropriate reference point recommendations. If the SYM should fail, then deterministic calculations using means, medians or trimmed means should replace the stochastic calculations. Should model steps prove to be inadequate, then a model-free approach should be used to determine an interim ABC.

References

Hart, D. R. (2013). Quantifying the tradeoff between precaution and yield in fishery reference points. *ICES Journal of Marine Science*, 70(3), 591-603.